

CLASSES AND OBJECT

The only difference b/w a structure and a class in C++ is that by default the members of a class are **private** while by the members of a structure are **public**.

```
#include <iostream>
```

```
using namespace std;
```

```
Class Complex
```

```
{
```

```
private:
```

```
int a, b;
```

```
public:
```

```
void set_data(int x, int y)
```

// Member function

```
{
```

```
    a = x;
```

```
    b = y;
```

```
}
```

```
void show_data()
```

```
{
```

```
    cout << "a =" << a << "b =" << b;
```

```
}
```

```
}
```

```
void main()
```

```
{
```

Complex C1; // Class ki madat se banne data type Complex Ka

C1.set_data(3,4); variable ko hum object kahte hai na ki

C1.show_data(); variable.

```
}
```



define

We can also declare Member function outside the class

Rules:

- Member function should be declare first inside the class.
- Between return type and function name we have to put | write membership label

→ then outside the class

Class name :: → Membership label

↓
scope resolution
operator

Important:

function defined inside class are by default **in-line function**



In C/C++ we can't directly add or use any other arithmetic operators between operand to get result if the variable are the declare by user-defined data-type.

By directly using arithmetic operators it will give errors.

```
#include <iostream>
```

```
using namespace std;
```

```
Class complex
```

```
{
```

```
private:
```

```
int a, b;
```

```
public:
```

```
void setData(int x, int y)
```

```
{
```

```
    a = x;
```

```
}
```

```
    b = y;
```

```
void showData()
```

```
{
```

```
    cout << "a = " << a << "b = " << b;
```

```
}
```

```
Complex add(Complex c)
```

```
{
```

```
    Complex temp;
```

$c1.a + c2.a$ **X** As $c1$ and $c2$ are define / declare under main function not here

$temp.a = a + c.a;$ Here we write directly a because

$temp.b = b + c.b;$ member function can access caller object directly

```
    return(temp);
```

```
}
```



void main()

{

complex c1, c2, c3;

c1.set_data(3, 4);

c2.set_data(5, 6);

c3 = c1.add(c2);

c3.show_data();

}

Here c1 is caller object and add() member function